Visualizing Honeypots Using Honeyd

Tom Chatto, Logan Flowers, Hailey Hudson, Tiffany Monarco

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1 Introduction

Hands-on experience is very important in cybersecurity education, however many students have limited access to tools where they can safely explore real-world concepts. Cybersecurity labs often require specialized hardware, extensive setup, and ongoing maintenance, which can make them difficult to deploy in classroom settings or for independent learners.

This project proposes the development of a containerized *Honeypot Lab*: a reproducible Docker-based environment designed to support hands-on cybersecurity education through cyber-deception concepts. The lab will use Honeyd, a low-interaction honeypot framework that simulates network stacks and services for virtual hosts, allowing users to observe and analyze attacker behavior in a controlled environment [6].

This project is meant to help students apply what they learn in cybersecurity courses through hands-on practice. The final result will be an easy-to-use lab where students can explore core cybersecurity concepts by experimenting with honeypots and analyzing network activity.

2 Research Context and Problem Statement

Cybersecurity is one of the most sought after fields today, with the employment of Information Security Analysts projected to grow 29% by 2034 [5]. With demand increasing, there must must also be a greater availability for education. However, a barrier we face is the gap between theoretical knowledge and applicable skills. How do we give students the experience and practice they need to translate their textbook knowledge to the skills they need in a career? Traditional Lab environments are difficult to maintain, expensive, and hard to scale. This renders this useful environment for students inaccessible. This is what our project aims to address. Our project focuses on Cybersecurity education with the Honeyd framework at the core of our research. We aim to utilize Honeyd to be able to offer hands-on Cybersecurity education in a controlled and accessible classroom environment.

2.1 Existing Approaches and Limitations

There are a number of existing approaches to the issues we want to address, though they each of their own limitations.

Commercial Platforms such as *CyberDefenders* [1] and *HackTheBox* [3] are challenge based learning environments and curricula offering deep cyber defense and analysis skills and certifications. Their limitations lie in price, with hundreds of dollars minimum for access to the course or certification exams.

Physical Lab Environments exist, which offer a hands-on physical laboratory for students to practice. Their limitations lie within a few reasons; They are specialized, expensive, need to be specially built and maintained, and rarely documented.

Docker-based labs, such as *Metasploit* [4] and *DVWA* [2], do teach vulnerability exploitation and pen-testing. Though, their limitations cap within that, they don't teach the defensive aspects of cybersecurity education.

These approaches fall short of the goals of this project because they either lack accessibility, lack critical aspects of education, or fall short of being approachable enough.

2.2 Problem Solving Statement

This project will address the following challenges:

1. Accessibility: Can we create a fully containerized honeypot environment that both a student and educator alike can deploy with a single command?
2. Education: Can we create a structured list of hands-on exercises with specific learning outcomes, packaging it within a curriculum?
3. Approachability: Can we provide approachable and comprehensive supporting materials for educators to be able to foster better learning?

3 Proposed Solution

To tackle the accessibility and hands-on learning challenges in cybersecurity education, this project will develop a containerized Honeypot Lab built around Honeyd. The lab will be packaged using Docker so users can quickly deploy the environment with minimal setup, making it easier for both students and instructors to use in classroom or independent learning settings.

The environment will include three preconfigured Honeyd profiles: a Windows XP host, a Linux SSH server, and a Linux web server. These profiles give students a more realistic environment to explore how different systems appear on a network while interacting with a low-interaction honeypot. Logging will capture activity and interactions in real time so students can review connection attempts and analyze attacker behavior.

To improve usability, the project will also include a simple web-based GUI for viewing logs and accessing challenges. Alongside the technical environment,

structured learning modules and hands-on exercises will be created to guide students through concepts such as through interacting with a honeypot, identifying a honeypot, and collecting data from Honeypots.

By combining an easy-to-deploy lab environment with educational content, this project will give students a hands-on way to apply cyber-deception concepts outside of lecture-based learning.

4 Evaluation and Implementation Plan

The Honeypot Lab will be evaluated based on both its technical functionality and how well it supports hands-on cybersecurity learning. Development will follow a phased approach focused on building the Docker environment, configuring the honeypot profiles, integrating logging and the GUI, and creating learning materials.

4.1 Evaluation Plan

This project will be evaluated in two areas: technical reliability and educational usability.

From the technical side, testing will focus on whether the Dockerized environment can be launched consistently with a single command. This includes making sure that Honeyd runs correctly, all three profiles behave as expected, and logging captures interactions in real time. The GUI will also be tested to ensure logs are displayed clearly and challenges are accessible.

From the educational side, users will complete hands-on challenges and user scenarios. Feedback will be collected on setup difficulty, challenge difficulty, and overall usability of the lab. The project will also be considered successful if users can set up the lab on their own, interact with the honeypot, and analyze logs using only the setup guides and documentation.

4.2 Timeline

The project will be completed in the following phases:

1. **Phase 1 (2/27/2026): Environment Setup and Dockerization:** Set up the base Docker environment and containerize Honeyd to ensure the lab can be launched consistently across systems.

2. **Phase 2 (3/13/2026): Honeypot Configuration and Logging** Configure the three Honeyd profiles: Windows XP, Linux SSH host, and Linux web server. Implement logging to capture and store network interactions.

3. **Phase 3 (3/27/2026): Curriculum and Challenge Design** Create learning modules, challenge prompts, expected outcomes, and solutions.

4. **Phase 4 (4/10/2026): Web GUI Development** Develop a simple web interface for viewing logs and accessing challenges.

5. **Phase 5 (4/24/2026): Testing and Evaluation** Test the full environment, gather user feedback, and refine the lab, documentation, and educational materials.

6. **Phase 6 (4/30/2026): Final Documentation and Deployment** Finalize the repository with setup instructions, README files, instructor materials, and learning modules.

5 Revisions

Revisions of previous drafts include rewriting the introduction to better include the problem statement and project description. The proposed solution was also revised to include more detail about the Honeyd profiles. Finally, the timeline was revised to more accurately track the progress of our work.

References

[1] CyberDefenders. Cybersecurity blue team labs & training, 2026.

[2] digininja. Damn vulnerable web application (dvwa), 2026.

[3] HackTheBox. Hack the box named a leader in the forrester wave for cybersecurity skills and training platforms, 2026.

[4] Metasploit. The world's most used penetration testing framework, 2026.

[5] US Bureau of Labor Statistics. Information security analysts. 2024.

[6] Niels Provos. A virutal honeypot framework. *Proceedings of the 13th USENIX Security Symposium*, 2004.